10-2 Data Analysis [1]

Point!

Data Organization

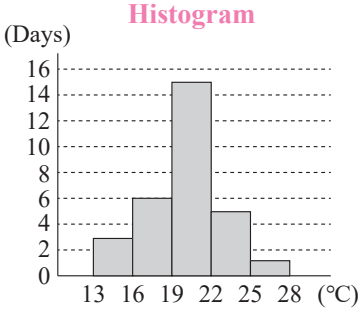
(1) ([1]**Frequency distribution table**) : A table that organizes and summarizes data, as shown on the right.

- The interval of the frequency distribution table is referred to as a ([2]**class**), the width of the interval is called the ([3]**class width**), and the midpoint value of the class is called the ([4]**class mark**).
- The number of data values included in each class is referred to as the ([5]**frequency**).

Frequency distribution table

Class (°C)	Frequency (days)
13 or higher and lower than 16	3
16 to 19	6
19 to 22	15
22 to 25	5
25 to 28	1
Total	30

Histogram



(2) ([6]**Histogram**) : A frequency distribution table presented as a bar graph. (see figure above)

(3) Measure of central tendency

Measure of central tendency : A single numerical value representing the characteristic of a dataset.

[1] ([7]**Mean**) : The value obtained by dividing the sum of the data by the number of data points. In spreadsheet software, it can be determined using the ([8]**AVERAGE**) function.

[Cell I3]=AVERAGE(B3:H3)

	A	B	C	D	E	F	G	H	I
1	Analysis of maximum temperatures in the 4th week of February 2023								
2		19th (Sun.)	20th (Mon.)	21st (Tue.)	22nd (Wed.)	23rd (Thurs.)	24th (Fri.)	25th (Sat.)	Mean
3	Tokyo	18.5	14.7	9.2	10.4	10.4	12.1	12.7	12.57
4	Osaka	14.3	12.7	8.5	10	12.5	8.5	10	10.93

[2] ([9]**Median**) : The value that appears at the center when all the data is arranged in ascending order. If the number of data points is even, the median is the mean of the two central values. In spreadsheet software, it can be determined using the ([10]**MEDIAN**) function.

[Cell I3]=MEDIAN(B3:H3)

	A	B	C	D	E	F	G	H	I
1	Analysis of maximum temperatures in the 4th week of February 2023								
2		19th (Sun.)	20th (Mon.)	21st (Tue.)	22nd (Wed.)	23rd (Thurs.)	24th (Fri.)	25th (Sat.)	Medium
3	Tokyo	18.5	14.7	9.2	10.4	10.4	12.1	12.7	12.10
4	Osaka	14.3	12.7	8.5	10	12.5	8.5	10	10.00

[3] ([11]**Mode**) : The value that appears most frequently. When arranging data in a frequency distribution table, the class mark of the class with the highest frequency is considered the mode. In spreadsheet software, it can be determined using the ([12]**MODE**) function.

[Cell I3]=MODE(B3:H3)

	A	B	C	D	E	F	G	H	I
1	Analysis of maximum temperatures in the 4th week of February 2023								
2		19th (Sun.)	20th (Mon.)	21st (Tue.)	22nd (Wed.)	23rd (Thurs.)	24th (Fri.)	25th (Sat.)	Mode
3	Tokyo	18.5	14.7	9.2	10.4	10.4	12.1	12.7	10.40
4	Osaka	14.3	12.7	8.5	10	12.5	8.5	10	8.50

Warm Up

Answer the following questions.

(1) The data on the right represents the results of a handball throw from a class of 32 students. Answer [1] and [2].

[1] Create a frequency distribution table where the class interval is 12 m to less than 15 m, with each class having a width of 3 m.

[2] Based on the frequency distribution table created in question [1], draw a histogram.

15	20	21	23	30	27	27	16
20	18	22	19	20	21	12	21
24	27	22	26	25	32	24	21
21	23	20	16	23	21	21	24

The unit is m.

(2) The data on the right represents the scores of 12 students who played a game. Determine the mean, median, and mode of the data.

6	9	3	6	10	12
4	3	8	10	7	6

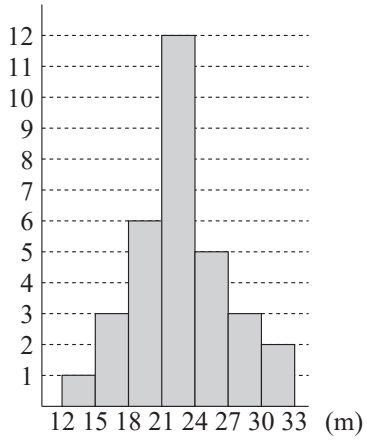
The unit is points.

Explanation

(1) [1]

Class (m)	Frequency (persons)
12 or higher and lower than 15	1
15 to 18	3
18 to 21	6
21 to 24	12
24 to 27	5
27 to 30	3
30 to 33	2
Total	32

[2] (Persons)



(2) Mean: $= \dfrac{6+9+3+6+10+12+4+3+8+10+7+6}{12} = \dfrac{84}{12} = 7$ 7 points

Median: First, arrange all the data in ascending order.

3, 3, 4, 6, 6, 6, 7, 8, 9, 10, 10, 12

Since the number of data points is even, the median is the mean of the two central values, which are 6 and 7.

Therefore, $\dfrac{6+7}{2} = 6.5$ 6.5 points

Mode: The value that occurs most frequently is 6 6 points

Try

1 The following data represents the quiz scores of 20 high school students. Answer the following questions.

3	4	9	7	6	10	5	5	7	9	
6	8	1	5	7	10	8	6	3	7	(Points)

(1) Determine the mean of the data.
(2) Determine the median of the data.
(3) Determine the mode of the data.

2 The following data shows the recorded daily maximum temperatures in Tokyo for February 2023. Answer the following questions.

13.1	9.2	6.2	11.2	13.6	15.4	11.7	10.6	3.5	12
14.1	16.9	10.3	10.7	7.8	9.6	10.8	15	18.5	14.7
9.2	10.4	14.4	12.1	12.7	10.7	19.4	15	(°C)	

(1) Create a frequency distribution table where the class interval is 14°C to less than 16°C, with each class having a width of 2°C.
(2) Based on the frequency distribution table created in step (1), draw a histogram.

3 The following table summarizes the functions available in spreadsheet software. Choose the term that best fits into the blanks [1] to [3] from the options **A** to **D** below, and answer using the letters.

Function	Explanation
([1])	Determines the mean.
([2])	Determines the median.
([3])	Determines the mode.

A MODE **B** AVERAGE **C** MEDIAN **D** VARP

Exercise

1 Answer the following questions.

(1) The following data records the heights of 30 students in a class. Answer the following questions.

168	151	168	166	183	162	167	176	161	175	
156	165	173	152	165	177	156	171	158	172	
172	148	146	167	173	155	169	180	161	163	(cm)

[1] Create a frequency distribution table where the class interval is 145 cm to less than 150 cm, with each class having a width of 5 cm.

[2] Based on the frequency distribution table created in question [1], draw a histogram.

(2) The following data represents the daily sleep duration of 20 high school students. Answer the following questions.

7	7	5	9	7	6	8	4	8	10	
3	8	6	8	9	5	8	6	8	8	(Time)

[1] Determine the mean of the data.

[2] Determine the median of the data.

[3] Determine the mode of the data.

2 Items **A** to **C** below describe functions available in spreadsheet software. Mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A The mean can be determined using the AVERAGE function in spreadsheet software.

B The median can be determined using the MEDIAN function in spreadsheet software.

C The mode can be determined using the SUM function in spreadsheet software.